

AI as Substrate Mediator: Paper 1’s Fourth Architectural Claim

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This work derives from and formalizes the Coordination Knowledge Substrate (CKS) pattern introduced in *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize Paper 1’s fourth architectural commitment — *AI as substrate mediator* — as a named, paper-level claim defined by five architecturally severable structural properties, and to map the foundational, operational, and anti-pattern sub-commitments that derive from it.

Abstract

The CKS pattern names *AI as substrate mediator* as one of its six architectural commitments; §4.2 of the source paper identifies it as the cross-claim spine connecting the substrate/LLM governance boundary to the multi-human coordination extension. This note formalizes the commitment as Paper 1’s Claim 4 with a five-property structural definition: the LLM (A) reads substrate as primary state, (B) writes under human-authored orchestration rules, (C) holds no substrate-relevant state outside the substrate, (D) exercises no authority over substrate content, and (E) has substrate-affecting writes recorded with attribution. The note’s central architectural move is to assert that these five properties are *severable*: each is independently testable, each is independently violable, and each violation produces a distinct failure mode. The mediator role is therefore not a single composite commitment but a conjunction of five separately defensible structural properties. The note states the claim, develops the severability structure, names the architectural position the role occupies (neither sovereign nor inert), maps three primary anti-patterns, traces the relationship to Paper 1’s Claim 1, enumerates derived sub-commitments, and provides a five-test operational verification procedure.

1. Why the mediator commitment needs a claim-level anchor

Paper 1 commits to six architectural claims, of which *AI as substrate mediator* is the fourth in the abstract’s enumeration. The commitment is named in §1, developed in §4.1–§4.2, and carried into the multi-human coordination extension at §9.1. A foundational derivation note (A1.04) already formalizes the five-property definition at sub-commitment level. That note develops the role; it does not anchor the role as Paper 1’s fourth named claim.

The anchor is needed for three reasons. First, the foundational note treats the mediator role as one architectural commitment among many; it does not assert the claim-

level status the source paper grants. Downstream prior-art work citing the mediator commitment as Paper 1’s Claim 4 needs an explicit landing site at that level. Second, the sub-commitment derivations developed in Series A — operational variants A2.18–A2.23 and anti-patterns A3.11–A3.13 — require a named claim-level parent; without one, the chain from paper-level claim through foundational sub-commitment to operational variant has a missing link. Third, and most consequentially for the prior-art posture, the severability of the five properties is itself a claim — not a property that follows automatically from naming the role. The severability move generates five separately defensible prior-art propositions plus the conjunction; without an anchor stating severability as part of the claim, those propositions lack a citable claim-level formulation.

2. The claim, stated

Paper 1’s fourth architectural claim is the following.

Claim 4 (AI as substrate mediator). In a CKS system, the LLM’s architectural role is *substrate mediator* between humans and substrate. The role is defined by five structural properties — substrate as primary state, writes under orchestration rules, no substrate-relevant state outside the substrate, no authority over substrate content, and substrate-affecting writes recorded with attribution — each of which is architecturally severable: independently testable, independently violable, and individually defensible as a structural commitment of the architecture.

The claim has three components. The *role assignment* is a positive specification, not a residual category: the LLM is a mediator, not an autonomous agent over substrate, not a source of truth, not a terminal producer, and not a passive substrate client. The *five-property definition* is the role’s operational content — the architecture commits to those five and only those five, and no proper subset suffices. The *severability assertion* states that the five are jointly definitive and individually independent: a system can satisfy four and violate the fifth, and the result fails Claim 4 in a specific, named way, not partially.

3. The five severable mediator properties

Each property states a structural commitment the architecture imposes on every LLM operation in the system, at all times during the system’s existence.

Property A — Substrate as primary state. The LLM reads from substrate content as its primary source of state for the operation it performs. In-context information not drawn from the substrate is admissible only insofar as it does not serve as a substitute for substrate state the operation depends on. The LLM’s parametric memory, prior-session context, and runtime scratchpad are not authoritative sources of substrate-relevant state.

Property B — Writes under orchestration rules. The LLM writes to substrate content only under orchestration rules authored by humans. The form of the rules is not specified; what is specified is that they are human-authored and the LLM operates

within them rather than authoring or modifying them. Writes outside any applicable orchestration rule are not authorized writes.

Property C — No shadow state. The LLM does not hold substrate-relevant state outside the substrate. Three forms of out-of-substrate state are ruled out: agent memory persisted across invocations and treated as authoritative, per-session storage that carries substrate-relevant state forward without being written into the substrate, and in-weight memory of substrate content treated as the answer to “what is the case.” Substrate-relevant state lives in the substrate.

Property D — No authority over substrate content. The LLM does not exercise authority over substrate content. It cannot silently overwrite content, collapse contradictions the substrate preserves, gate human access, refuse authorized writes, or modify the orchestration rules under which it operates. Authority sits with humans, per the human-governed commitment (Paper 1’s Claim 3); the LLM executes within the boundaries that authority defines.

Property E — Substrate-affecting writes recorded with attribution. When an LLM operation results in a write to substrate state, the write is itself recorded in the substrate as substrate content, with attribution sufficient to identify it as LLM-authored and to associate it with the orchestration rule under which it was produced. There are no unattributed substrate changes from the LLM, and no LLM outputs that affect substrate state without being recorded in the substrate.

The five trace to specific commitments in the source paper. Properties A and B follow from §4.1’s substrate/LLM governance boundary; C follows from §11.3’s source-of-truth commitment for the substrate; D follows from the human-governed commitment §3.3 carries; E follows from the path-retraceability commitment §3.1 develops.

4. Severability: independent testability and independent violability

Severability has two operational components.

Independent testability. Each property admits a binary test that is decidable for any given LLM operation without reference to the other four. The five tests are listed in §9.

Independent violability. A system can satisfy four properties and violate the fifth, and the result is identifiable as a specific failure of Claim 4 rather than a general failure of the mediator role. Each violation produces a distinct architectural failure mode with a distinct remediation path: a system that violates only Property C (holds shadow state) is not the same architectural defect as one that violates only Property D (exercises authority), and naming the violation precisely is what makes the fix possible.

Severability generates the prior-art structure the anchor commits to. Each of the five properties is independently defensible — five separately citable propositions. The conjunction of all five, with severability asserted, is the sixth proposition: that the role is exhausted by exactly these five and no proper subset suffices. A subsequent system that claims the mediator role as novel must satisfy a definition equivalent to all five under some renaming, name a sixth property the architecture omits, or defend

a subset as sufficient — none of which is open without engaging the five-property anchor.

5. The architectural position: between sovereign and inert

The mediator role is a specific position between two adjacent positions other AI-system architectures occupy. Naming it precisely keeps the role from being read as a midpoint compromise or a rebranding of either neighbor.

Not sovereign. A sovereign LLM holds substrate-relevant state in its own memory, treats it as authoritative, decides what counts as substrate truth, and exercises authority over substrate content. This is the LLM-as-agent and LLM-as-source-of-truth posture; it violates Properties A, C, and D. The mediator role cannot be reached by softening sovereignty with policy promises — it is *architecturally* non-sovereign: the LLM does not hold the authority, structurally, that a sovereign role would require.

Not inert. An inert LLM is a passive producer — generates outputs in response to prompts, terminates, and the substrate, if any, does not record what the LLM produced or what reasoning it performed. This is the LLM-as-terminal-producer posture; it violates Property E and frequently Property A.

The mediator position is neither. It is *active without being sovereign*: the LLM reads, reasons, drafts, interprets, and writes — but reads from substrate, writes under rules, holds no shadow state, exercises no authority, and produces records the substrate carries. The five properties define exactly that position — not “less sovereignty” and not “more activity,” but a third architectural option specified by the conjunction, precise enough that no soft formulation of sovereignty or inertness lands on it.

6. Anti-patterns the claim rules out

The five-property definition rules out three primary anti-patterns. Each is treated as a standalone derivation note in Phase A3; the claim-level anchor maps them.

LLM as autonomous agent (A3.11). Holds goals, plans, and intermediate state across multi-step operations; exercises judgment about substrate writes. Violates Property D (agent authority over substrate), Property B (writes outside orchestration rules), and frequently Properties A and C (state held in agent memory rather than read from substrate).

LLM as terminal producer (A3.12). Generates outputs as deliverables in their own right; outputs are not recorded as substrate content. Violates Property E. Common in conventional LLM applications where the substrate, if any, is incidental scaffolding rather than the primary artifact; CKS reverses the asymmetry.

LLM as source of truth (A3.13). Treats the LLM’s parametric memory, prior context, or runtime state as the authoritative answer to “what is the case.” Violates Properties A and C. The pattern is what §6.2 of the source paper names as *context rot*: capacity overflow, compaction loss, and goal drift under repeated compression of state the substrate could have held instead.

7. Relationship to Paper 1’s Claim 1

Paper 1’s Claim 1 — the hybrid substrate/LLM commitment — establishes that CKS systems consist of a substrate and an LLM operating together, with a division of labor at the governance boundary. Claim 1 names the existence of both elements; it does not specify what the LLM’s side of that division consists of, structurally.

Claim 4 supplies that specification. Without it, “the LLM does the high-dimensional reasoning” leaves the LLM’s authority over substrate content, its state management, and the recording of its outputs underspecified. With Claim 4, the LLM’s role is closed: it reads substrate, writes under rules, holds no shadow state, exercises no authority, and produces records. The hybrid Claim 1 commits to is a hybrid of substrate plus *governed mediator*, not a hybrid of substrate plus LLM-of-undefined-authority. The two claims compose: Claim 1 without Claim 4 has not specified an architecture; Claim 4 without Claim 1 has not specified what the LLM is mediating between.

8. Derived sub-commitments

The Claim 4 anchor parents the following Series A derivations:

- **A1.04** — Foundational derivation of the mediator role as a single architectural commitment with the five-property definition. (Existing.)
- **A2.18** — Severability as architectural decomposition: formalizes the independence of the five properties as a structural claim, not merely a property of the foundational definition.
- **A2.19** — Property A as standalone sub-commitment: substrate as primary state.
- **A2.20** — Property B as standalone sub-commitment: writes under human-authored orchestration rules.
- **A2.21** — Property C as standalone sub-commitment: no substrate-relevant state outside the substrate.
- **A2.22** — Property D as standalone sub-commitment: no authority over substrate content.
- **A2.23** — Property E as standalone sub-commitment: substrate-affecting writes recorded with attribution.
- **A3.11** — Anti-pattern: LLM as autonomous agent.
- **A3.12** — Anti-pattern: LLM as terminal producer.
- **A3.13** — Anti-pattern: LLM as source of truth.

The Phase A2 sub-commitments are operational decompositions: each takes one property and formalizes it as a standalone commitment with its own architectural content, failure modes, and operational test. The Phase A3 anti-patterns are the negative space: each names a pattern the claim rules out and identifies the properties violated. Together, five positive sub-commitments and three anti-patterns give the Claim 4 anchor its full derivation tree at the foundational and operational levels.

9. Operational test

A system satisfies Paper 1’s Claim 4 if and only if all of the following five tests return Yes for every LLM operation in the system, at all times during the system’s existence:

1. **Property A test.** Did the LLM read from substrate content as its primary source of state for the operation? Yes / No.
2. **Property B test.** Did the LLM write to substrate content only under an applicable orchestration rule authored by humans? Yes / No.
3. **Property C test.** Did the LLM hold no substrate-relevant state outside the substrate during the operation — no agent memory across invocations treated as authoritative, no per-session storage of substrate-relevant state, no in-weight memory of substrate content treated as the answer to “what is the case”? Yes / No.
4. **Property D test.** Did the LLM exercise no authority over substrate content — no silent overwrite, no collapse of preserved contradictions, no gating of human access, no refusal of authorized writes, no modification of orchestration rules? Yes / No.
5. **Property E test.** Was every substrate-affecting result of the LLM operation recorded in the substrate as substrate content, with attribution identifying it as LLM-authored and associating it with the orchestration rule under which it was produced? Yes / No.

A system that returns *Yes* to all five satisfies Claim 4. A system that returns *No* to any one fails Claim 4 in a specific, identified way: the failing test names the property violated, the violation maps to a specific anti-pattern, and the remediation path is to bring the failing property into compliance without disturbing the four already satisfied. Per the severability structure of the claim, partial satisfaction is not partial compliance — it is failure with a named cause.

10. Conclusion

Paper 1’s Claim 4 commits the architecture to a specific role for the LLM in a CKS system: substrate mediator. The role is defined by five architecturally severable structural properties — substrate as primary state, writes under orchestration rules, no shadow state, no authority over substrate content, and substrate-affecting writes recorded with attribution. Each is independently testable and independently violable; each is individually defensible as a structural commitment; the conjunction is the definitional claim that the role is exhausted by these five and no proper subset suffices.

The claim sits between two adjacent positions other AI-system architectures occupy. It is not sovereignty under softer rhetoric; it is not inertness with more activity. It is a third position specified by the five properties, and the five are precise enough that the position is architectural rather than rhetorical.

Subsequent work that adopts the CKS pattern, extends it, composes it with adjacent patterns, or argues against it should treat *AI as substrate mediator* as the role formalized here. Subsequent work that grants LLMs authority over substrate content, holds substrate-relevant state outside the substrate, produces outputs that affect substrate state without recording them, or writes outside orchestration rules is using a different architectural role, and the difference should be named at the level of the specific property the difference touches.

Source paper

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

How to cite this note

Li, W. (2026). *AI as Substrate Mediator: Paper 1's Fourth Architectural Claim*. Derivation note #421 (A0.04). May 14, 2026. ORCID: 0009-0004-8065-3235.